MILITARY SPECIFICATION

CLOTH, KNITTED, POLYAMIDE, HIGH TEMPERATURE RESISTANT, REVERSE JERSEY

1. SCOPE

1.1 This specification covers the requirements for one type of knitted fabric of a high temperature resistant, aromatic polyamide for use as seat cushion covers in aircraft.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

PPP-P-51	Packaging, Packing, and Marking of Textile Fabrics (Woolens, Worsteds, Cotton, Silks and Synthetics)

STANDARDS

Federal

Fed Std No. 191	Textile Test Methods

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions shall be obtained from the procuring activity or as directed by the contracting officer.)

FSC 8305

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3. REQUIREMENTS

3.1 Standard sample. The cloth shall match the standard sample for shade and shall be equal to or better than the standard sample with respect to all characteristics for which the standard sample is referenced.

3.2 First article. This specification contains provisions for first article inspection and approval (see 4.2, 6.2, and 6.6).

3.3 Material

3.3.1 Yarn. The yarn used in the manufacture of the knitted cloth shall be a bright, non-melting, aromatic polyamide solution dyed Sage Green color 1565.

3.3.2 Denier and filament. The yarn used shall be 200 denier 100 filament.

3.3.3 Twist. The yarn used for both beams shall be twisted to a maximum of 3 turns per inch Z.

3.4 Physical requirements. The finished cloth shall conform to the requirements specified in table I when tested as specified in 4.4.

3.5 Width. Unless otherwise specified, the overall minimum width shall be as specified by the procuring activity.

3.6 Finish. The cloth shall be finished as specified in 6.5, including an antistatic finish if required by the procuring activity.

Table I. Physical properties

Property	Test Method or Paragraph	Minimum
Construction (wales X courses)	5070	24 X 30
Weight (ounces/yard ^{2})	5041	7.0
Abrasion (cycles to destruct)	5302 1/	1,000
Bursting strength ball method (lbs.)	5120	150+ (2/)
Stretch (% wales X courses)	4.4.1	35 (maximum)
		70 (maximum)
Set (% wales X courses)	4.4.1	5.0 (maximum)
		10.0 (maximum)

1/ "0" grit emery paper, 3 psi diaphragm pressure, 2 lb. load.

2/ Fabric does not burst at 150 lbs., maximum on Scott Tester.

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3.7 Workmanship. The cloth shall conform to the quality and grade of product established by this specification. The occurrence of defects shall not exceed the acceptable quality levels specified herein (see 6.4).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Certificate of compliance. Where certificates of compliance are submitted the Government reserves the right to check test such items to determine the validity of the certification.

4.2 First article inspection. The preproduction sample submitted in accordance with 3.1 shall be visually inspected for appearance, color and finish. The sample shall be tested for physical properties in accordance with the methods specified in 4.4.

4.3 Sampling and acceptability criteria. Quality conformance inspection samples shall be taken from each lot of cloth. A lot shall consist of cloth made under essentially the same conditions and in a single production run. Unless otherwise specified, the lot size shall be expressed in units of 1 yard each.

4.3.1 Samples for yard-by-yard examination. The sample unit for this examination shall be 1 linear yard. A sample size for the yard-by-yard examination shall be selected in accordance with MIL-STD-105, Inspection Level II. The required sample yardage to be examined shall comprise an approximately equal number of yards out of each selected roll.

4.3.2 Sample for overall and dimensional examination. The sample unit for these examinations shall be one roll. Examination shall be in accordance with MIL-STD-105, Inspection Level S-2, Acceptance Number zero. The minimum sample size shall be three rolls.

4.3.3 Samples for tests. The sample unit shall be 5 yards. The sample shall be in accordance with the following:

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Table II. Sample size

Lot Size (Yards)	Sample Size (Units)
800 or less	3 Min.
801 to 22,000 inclusive	5 Min.
Over 22,000	8 Min.

The lot shall be unacceptable if one or more units fail to meet any requirement specified.

4.3.4 In-process material examination. The examination will be made at any point or any phase of the manufacturing process to determine whether the material, construction and workmanship is as specified in Section 3.

4.3.5 Yard-by-yard examination. In each piece of cloth selected, an approximately equal number of yards comprising the sample shall be subjected to the examination. The defects found shall be identified and classified in accordance with table III. The defects found shall be counted regardless of their proximity to each other except where two or more defects represent a single local condition of the cloth, in which case only the more serious defects shall be counted. A continuous defect shall be counted as one defect for each wale-wise yard or fraction thereof in which it occurs.

4.3.6 Overall examination. The required number of pieces of finished cloth in the sample (see 4.3.2) shall be examined for defects as listed in table IV. Each defect shall be counted no more than once in each piece examined.

4.3.7 Length of individual rolls examination. The rolls shall be examined for gross length. Any gross length found to be less than the specified minimum length, or any gross length found to be more than 2 yards below the gross length marked on the roll ticket, shall be considered as a defect with respect to length. The sample unit shall be one roll. The number of rolls selected as a sample and the acceptance number for this examination shall be in accordance with table II.

4.3.8 Total yardage in sample examination. The lot size shall be unacceptable if the total of the actual gross lengths of rolls in the sample is less than the total of the gross lengths marked on the roll tickets.

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Table III. Classification of defects

Defect	Classification Major	Minor
Hole, cut, tear, any.	X	
Run or dropped stitch, end out.	X	
Tuck stitch, pulled stitch or needle streak.		X
Mend.	X	
Abrasion mark resulting from weak area.1/	X	
Thin or uneven cloth.1/	X	
Course or fine yarn.1/		X
Tight twist yarn.		X
Slubs or gouts, more than twice the size of normal yarn.		X
Knots extending above the surface of cloth.		X
Kinks or untrimmed ends.		X
Crease or wrinkle, (embedded).		X
Stretch (stitch tension) knitting tension too tight or too loose.	X	
Spot or stain.	X	

1/ Clearly visible at normal inspection distance (3 feet).

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Table IV. Overall defects

Defects
Objectionable odor.
Overall uncleanness.
Uneven knitting throughout the roll.
Edges frayed, ravelled, curled, rolled, folded, scalloped or uneven continuously or intermittently throughout the roll.
Width - less than minimum or more than maximum specified.

4.4 Tests. The methods of testing specified in Fed Std No. 191 wherever applicable, as listed in table V, and the following subparagraphs shall be followed. The physical and chemical values specified in Section 3 apply to the average of the determinations made on a unit of product for test purposes as specified in the applicable test methods. Failure of any sample of cloth to conform to the requirements specified in Section 3 shall be cause for rejection of the lot represented.

Table V. Test methods

Characteristics	Test Method (Fed Std No. 191) or paragraphs
Construction (wales and courses)	5070
Weight (ounces/yard^2)	5041
Strength bursting (lbs.)	5120
Abrasion (cycles to destruct)	5302
Stretch (%)	4.4.1
Set (%)	4.4.1

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4.4.1 Test for stretch

4.4.1.1 Test specimen. The test specimen shall be a rectangle 3 inches by not less than 9 inches. The long dimension shall be parallel to the length (wales) of the cloth for length tests and parallel to the width (courses) for width tests. The test specimen shall be taken no nearer the selvage than 1/10 the width of the cloth. Two parallel lines, 3 inches apart shall be drawn at right angles to and centered on the long dimension of the specimen. In addition, a line shall be drawn parallel to the long dimension of the specimen and 1 inch in from either edge for properly aligning the specimen in the jaws.

4.4.1.2 Test apparatus. The test device (figure 1) shall be an apparatus wherein the specimen is held between two clamps. The design of the upper clamp shall be such that one gripping surface or jaw shall be an integral part of the rigid frame of the clamp. The face of the movable jaw shall measure 1 inch by 5 inches minimum with the long dimension perpendicular to the direction of the application of the load. The lower clamp shall have similar jaws 2 inches by a minimum of 4 inches and when the weight is attached, shall weigh a total of 27 pounds. The jaws shall have smooth gripping surface, sufficiently flat and parallel to prevent slipping of the specimen during the test.

4.4.1.3 Procedure. The specimen shall be placed symmetrically in the clamps of the apparatus with the long dimension parallel and the short dimension at right angles to the direction of the application of the load. Using the specified guidelines, the specimen shall be aligned the outside edge of each jaw shall be 1 inch from the outside edge of the jaw. The proper weight shall be added to insure a total of 27 pounds (including the weight of the lower jaws). The load shall be applied for 10 minutes and the distance between the jaws measured by a pair of dividers or other suitable instrument. The specimen shall then be removed from the apparatus and allowed to relax for 10 minutes. The marks shall again be measured and the percent stretch and set shall be calculated as follows:

$$\text{Percent stretch} = \frac{\text{Stretched dimension in the apparatus minus the original 3 inch dimension}}{\text{The original 3 inch dimension}} \times 100$$

$$\text{Percent set} = \frac{\text{Relaxed dimension after 10 minutes minus original 3 inch dimension}}{\text{Original 3 inch dimension}} \times 100$$

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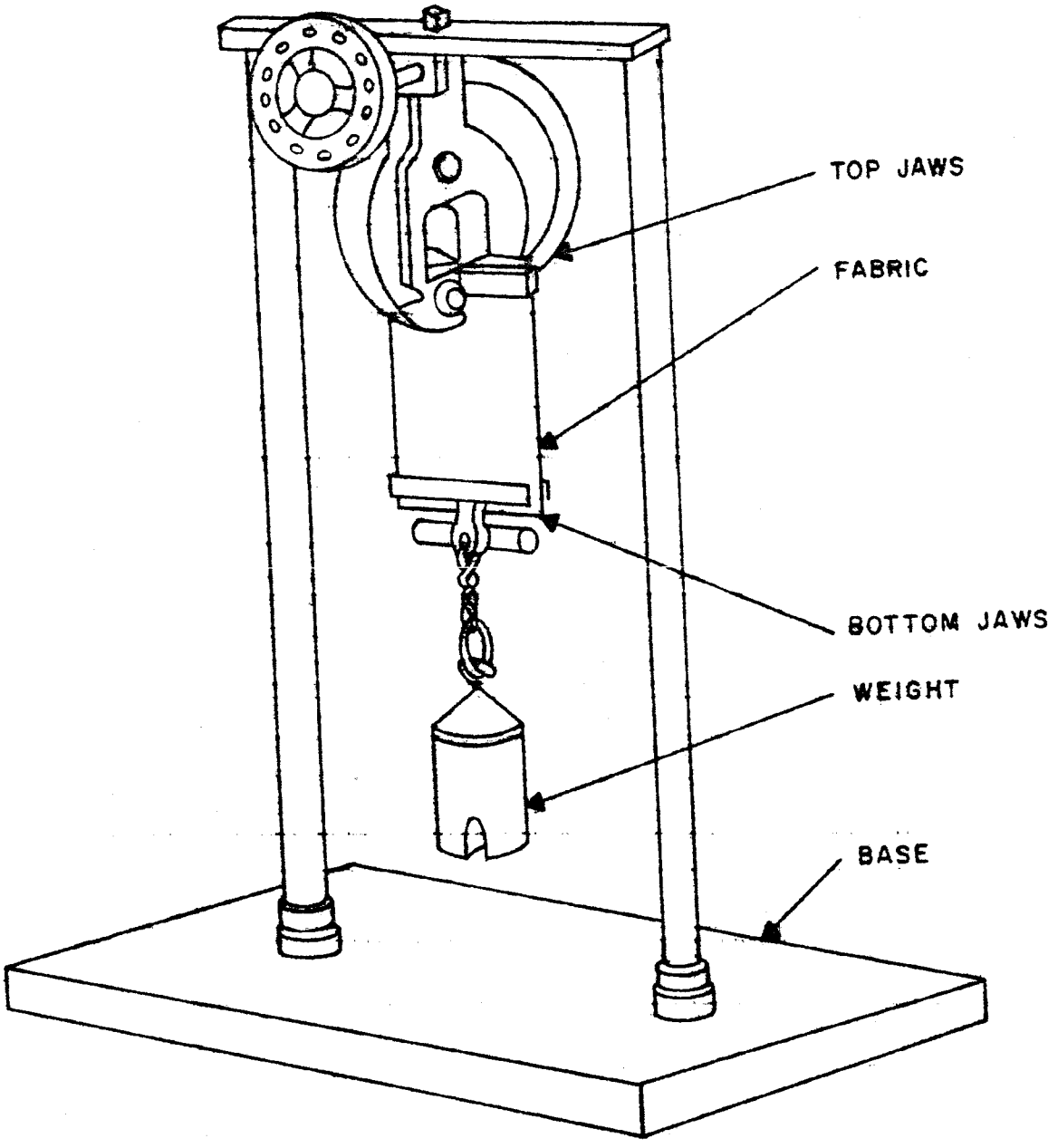


FIGURE 1. FABRIC STRETCH APPARATUS

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4.4.1.4 Report. One specimen each from the length (wales) and from the width (courses) direction shall be tested from each sample unit and the results obtained reported separately to the nearest percent. For instance, set (percent) wales X courses and stretch (percent) wales X courses.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking. The cloth shall be packaged, packed and marked in accordance with PPP-P-51. The levels of packaging and packing to be used in the preparation for delivery shall be as specified by the procuring activity.

6. NOTES

6.1 Intended use. This cloth is intended for use in the manufacture of seat cushions for use in aircraft.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.

b. Dimensions of cloth required (see 3.4).

c. Selection of applicable levels of packaging and packing (see 5.1).

d. First article (see 3.2).

6.2.1 In the preparation of contracts and orders, it must be noted that various options, choices and alternatives as indicated in PPP-P-51, may be exercised in the preparation of this cloth for delivery.

6.3 Knitting machine settings. The following is for information only, and is not a mandatory requirement. The cloth has been made on a 2-bar 20 gauge tricot knitting machine. The knitting quality should be 14 inches to a rack (480 courses). The knitting pattern shall be as follows:

Bar No. 1 - 1/2, 1/0
Bar No. 2 - 1/0, 2/3
Runner length:
  Bar No. 1 - 82.25" per rack (480 courses)
  Bar No. 2 - 101" per rack (480 courses)
Ratio:
  Front bar - 1.0
  Back bar - 1.225

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6.4 Defect level. Because of problems in producing aromatic polyamide continuous filament yarns, a special defect level will be allowed. Defects caused by broken filaments should not exceed 1 (one) per 4 (four) running yards. The supplier should make all efforts to improve the defect level through yarn inspection before and during warping.

6.5 Finishing procedure. The following is for information only, and is not a mandatory requirement. Finishing of the fabric should follow this procedure:

a. Scour - Open width, 160°-212°F using 1.0 g/l TSP and a detergent.

b. Dry and set construction at 300°F (Construction 28 X 33 or higher).

c. Autoclave heat set on roll
5 minutes pre-vacuum
30 minutes 28 psi steam
5 minutes vacuum
30 minutes 28 psi steam
10 minutes vacuum.

d. Antistatic treatment (optional)
6.6 percent Aston 123*
2.7 percent Accelerator EN
Set pH at 6.5
Pad on 50-60 percent wet pickup
Dry at 300°F
Cure at 350°F for 1.5 minutes.

(Note: Construction will relax during autoclaving so frame should be overfed.)

* The requirements for the durable antistatic treatment can be met with Aston 123, Onyx Chemical Company, 190 Warren Street, Jersey City, New Jersey 07032. Other products considered for this use must have prior approval of the contracting officer.

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6.6 First article. The cloth covered by this specification requires first article inspection and approval under the appropriate provisions of paragraph 7-104.55 of the Armed Services Procurement Regulations. The first article should consist of 5 yards of the dyed and finished cloth and shall be submitted or made ready for the contracting officer or his authorized representative. The contracting officer should include specific instructions in all procurement instruments, regarding arrangements for examination, test and approval of the first article.

Custodian:
Air Force - 11

Preparing activity:
Air Force - 11

PROJECT NUMBER: 8305-F237

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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